



Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LAB 12 INSTRUCTIONS

For

Interpret echo and unreachable messages

Course Code: C1123

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 5.1

Lab 12 — Interpret echo and unreachable messages

Course: Wireshark Network Analysis Masterclass (C1123)

Topic: 12 — ICMP Traffic Analysis

Learning outcome: LO3 — Interpret DNS, ARP, IPv4, ICMP and UDP evidence.

Time: about 30 minutes

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Scenario

Ping to the server works, yet a diagnostic tool on UDP port 9999 fails. You use ICMP evidence to explain the difference.

Goal

Echo requests/replies show an IP exchange; they do not guarantee an application works.

What you will produce

An ICMP event log explaining the service refusal.

Before you start

- Wireshark 4.6 or later, with the TShark command-line tools on PATH.
- Python 3 for the verification and export scripts (Windows: use `py -3` in place of `python3`).
- Open this lab folder in a terminal; every file the lab needs is inside it.
- Use only the supplied synthetic captures — do not capture on a network you are not authorised to monitor.

Files for this lab

File	What it is for
data/branch-office.pcap	Synthetic branch-office capture used by the lab
assets/scenario.md	The help-desk ticket that sets the scenario
assets/checks.json	Filters and frame counts the fixture must satisfy
assets/expected-evidence.png	The packet list your filter should produce
scripts/verify.py	Checks the capture facts with TShark
scripts/export_evidence.py	Exports this lab's evidence rows to outputs/evidence.csv
outputs/findings.md	Findings sheet you complete
assets/icmp-events.csv	ICMP event log you complete

Lab at a glance

1. Pair echo requests and replies
2. Read the quoted headers in the unreachable
3. Find the original UDP datagram
4. Record the events and your conclusion

Step-by-step

1. Prepare your lab workspace. Open this lab folder in a terminal. Windows uses `py -3` in place of `python3`. Wireshark GUI alone is sufficient for the investigation; CLI scripts require TShark on PATH. Run the fixture verification first.

```
python3 scripts/verify.py
```

2. Open `data/branch-office.pcap`. Apply `icmp.type == 8 || icmp.type == 0`. Pair echo sequences 0, 1 and 2 using identifier 7. Measure 0.030 seconds per pair.

3. Apply `icmp.type == 3 && icmp.code == 3`. Expand the quoted IPv4 and UDP headers. Record destination UDP port 9999.

4. Apply `udp.dstport == 9999` to locate the initiating datagram. Record the difference between a UDP service refusal and an echo reply.

5. Complete `assets/icmp-events.csv` with type, code, quoted protocol, port and conclusion. State which evidence supports a service refusal.

6. Export a reproducible packet table. Record the filter, frame numbers, measurement, explanation and limitation in `outputs/findings.md`.

```
python3 scripts/export_evidence.py
```

Expected evidence

With the lab filter applied, your packet list should match the frames below (produced by TShark from this lab's own capture).

icmp.type == 3 && icmp.code == 3					
No.	Time	Source	Destination	Protocol	Info
16	1.020000	192.0.2.20	192.0.2.10	ICMP	Destination unreachable (Port unreachable)
branch-office.pcap Displayed: 1					

Lab 12 expected evidence

Test it

The supplied `icmp.type == 3 && icmp.code == 3` expression matches 1 frame(s) in the specified capture. Run `scripts/verify.py` and compare the listed frame numbers. Keep your findings and exported table in `outputs/`. Explain the observed result rather than only copying a count.

Troubleshooting

- TShark not found: install Wireshark CLI tools and add the installation folder to PATH; on Windows use the Wireshark install directory.
- A filter returns zero: clear other filters, use the specified capture, and check the expression is in the display toolbar.
- TLS remains opaque: select the matching lab-tls.keys file by absolute path, reload, and remove a key-log preference from another lab.

Try it with TShark

The same evidence from the command line — run it from this lab folder:

```
tshark -n -r data/branch-office.pcap -Y icmp -T fields -e frame.number -e icmp.type -e icmp.code -e icmp.seq
```

Challenge

Develop and explain an alternative filter.

Reflection

Which second observation point would strengthen your conclusion?

Extension (optional)

ICMP lab — identify Time Exceeded (type 11) messages in the traceroute trace. Source: J.F. Kurose and K.W. Ross, Wireshark Labs (gaia.cs.umass.edu/kurose_ross/wireshark.php).

Reset

Clear all display filters and return to the C1123-Analyst or Default profile. If you loaded a TLS key log, remove it from Preferences > Protocols > TLS. The supplied captures never need regenerating for the core lab.

Note: The same steps appear in the Learner Guide. The slides show only the scenario and a summary.